Problem 1: (5 points)

Consider the undirected graph in Figure 1. What are the partitions identified by the Girvan-Newman algorithm after 1, 2, 3, 4, and 5 steps? Please explain your answer.

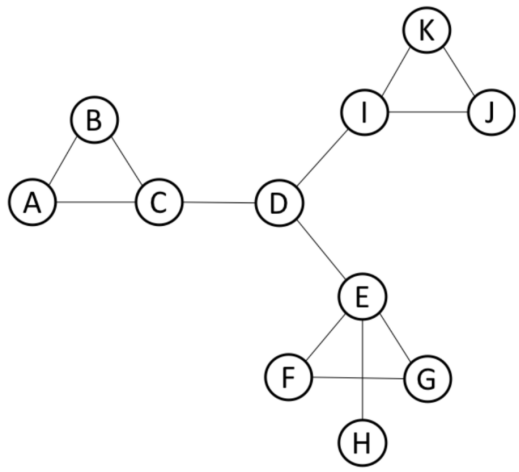

Figure 1: Identify the partitions in this graph using the Girvan-Newman algorithm.

Problem 2: (5 points)

For this problem, you will embed the network structure into the Euclidean space and train machine learning models on the embedding and properties of the network.

1. Download the three network datasets from the course Moodle page. Each dataset consists of a graph $G = (V, E)$ in the form of the adjacency matrix $\mathbf{A} \in \{0, 1\}^{n \times n}$ (`graph_<number>_A.pkl`) where $n = |V| = 200$, and the node attribute matrix $\mathbf{X} \in \mathcal{R}^{n \times 2}$ (`graph_<number>_X.pkl`). The i-th row of $\mathbf{X}$ corresponds to the attribute values for node i and the corresponding i-th row in $\mathbf{A}$ represents its adjacency information. Additionally, there is a vector $\mathbf{Y} \in \{0, 1, 2, 3\}^n$ (`graph_<number>_Y.pkl`) representing the corresponding labels of the nodes.

2. Compute the unnormalized graph Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{A}$ for each graph. Compute Laplacian eigenvectors $\mathbf{u}_2$ and $\mathbf{u}_3$ corresponding to the second and third smallest (in magnitude) eigenvalues of the Laplacian, respectively. *Tip*: You are free to use any method for this, but you may find scipy.sparse.linalg.eigsh useful.

3. Draw one figure for each network dataset with 2 subplots. Each subplot should be a two-dimensional scatter plot that contains one point (dot) for each node. For the first subplot, the two axes correspond to the two node attributes ($\mathbf{X}$). For the second subplot, the two axes correspond to the two eigenvectors ($\mathbf{u}_2$, $\mathbf{u}_3$) – this is the spectral embedding of the corresponding network. Color the points with different colors, according to the corresponding labels $\mathbf{Y}$.

4. Using NetworkX and Matplotlib, visualize the network using the `spectral` and `spring` layouts and provide the visualization as figure. Briefly compare with the spectral embedding $[\mathbf{u}_2\, \mathbf{u}_3]$.

5. For each dataset, train one Random Forest model on each for the following feature matrices (a) $\mathbf{u}_2$, (b) $\mathbf{u}_3$, (c) $[\mathbf{u}_2\, \mathbf{u}_3]$, (d) $\mathbf{X}$, and (e) $[\mathbf{X}\, \mathbf{u}_2\, \mathbf{u}_3]$ to predict label $\mathbf{Y}$. Use 40% of the dataset for training and the rest for testing. (You may want to use the `scikit-learn` library in Python.) Make sure to use the same train-test split for each of the five models.

6. For each dataset, draw one barplot where the X-axis denotes the five Random Forest models described above and the Y-axis denotes the classification accuracy on the test set. Make sure to define appropriate legends. Compare the performance between the models: which models perform better; does the observed performance match what you would expect based on the available information (before training the models)?

Instructions:

Submit a zip file that contains the following required files:

- `solutions.pdf`: Writeup of your answers.
- `asg4_p2.py` or `asg4_p2.ipynb`: Code-file(s) for data analysis and plots.

For purposes of reproducibility, make sure to set a seed for the random operations in your code.